

ACSS-PSL at #SMM4H-HeaRD 2025: Multilingual and Monolingual Adverse Event Classification using Transformers (Tasks 1 & 6)

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Introduction

Formal reporting systems from regulatory agencies like the FDA or EMA are central to pharmacovigilance but are known for significant under-reporting. In parallel, patients frequently use social media platforms like Reddit and X to share personal experiences with medications. This user-generated content forms a large body of real-world data offering valuable insights, such as detecting novel safety signals or understanding patient perspectives. However, the informal language, high volume, and multilingual nature of this data require advanced Natural Language Processing (NLP) techniques for effective analysis. Our work tackles two such tasks from the 2025 #SMM4H Shared Tasks.

Tasks Description

Task 1: Multilingual ADE Classification

This task required identifying social media posts mentioning Adverse Drug Events (ADEs) across English, French, German, and Russian. A post is labeled positive if it contains a personal mention of an ADE. The dataset was highly imbalanced, with the positive class being rare.

Table 1. Task 1: Train and Dev Set Distribution by Language

Language	Train Total	Train Pos (%)	Dev Total	Dev Pos (%)
German (de)	1,482	83 (5.6%)	634	35 (5.5%)
English (en)	17,974	1,214 (6.8%)	902	61 (6.8%)
French (fr)	977	70 (7.2%)	419	30 (7.2%)
Russian (ru)	10,754	1,084 (10.1%)	2,670	272 (10.2%)
All	31,187	2,451 (7.9%)	4,625	398 (8.6%)

Task 6: Shingles VAEM Classification

This task involved identifying Reddit posts reporting personal mentions of Shingles vaccine adverse events or vaccine failures (VAEM). Posts were labeled positive for first-hand experiences of an adverse reaction to the Shingrix vaccine or its failure. The dataset was relatively balanced, with the positive class representing about 46% of the data.

Implementation Details

The code for our experiments is publicly available on GitHub. All models were trained using the PyTorch framework on a single NVIDIA RTX 3070 GPU with 8GB of VRAM.

Discussion & Conclusion

Our work demonstrates the effectiveness of transformer models on noisy social media data, even without any text preprocessing.

- **Task 1 (Multilingual):** Our XLM-ROBERTa-Large model achieved an overall F1-score of 0.633. Performance varied significantly by language (F1 > 0.72 for English/French, vs. 0.54 for Russian), likely due to the heterogeneity of data sources.
- **Task 6 (Shingles Vaccine):** Our DeBERTa-V3-Base model with 5-fold cross-validation achieved a high F1-score of 0.951. This result highlights the model's robustness to noisy content, suggesting its pre-training was sufficient to handle social media specificities like URLs or informalities.
- **Key Strategy:** Class weighting was a crucial technique to handle data imbalance in both tasks.

System Configuration

For both tasks, no text preprocessing was performed. A class-weighted cross-entropy loss was used to mitigate data imbalance. The detailed hyperparameters are compared below.

Table 2. Comparison of training configurations.

Hyperparameter	Task 1 (Multilingual ADE)	Task 6 (Shingles VAEM)
Model	XLM-ROBERTa-Large	DeBERTa-v3-base
Optimizer	AdaFactor	AdamW
Learning Rate	1e-5	2e-5
Weight Decay	0.05	0.01
Epochs	6	Up to 6
Effective Batch Size	32	8
Gradient Accumulation	Yes	No
Learning Rate Scheduler	Cosine	Default
Warmup Ratio	10%	None
Cross-Validation	None	5-fold stratified
Ensembling	None	Averaged probabilities
Dropout	0.3	Default
Max Sequence Length	256 tokens	512 tokens

Results

Task 1: Overall F1-Score of 0.633

Our system consistently outperformed the participant mean and median across all metrics and languages, with particularly strong performance in French and English.

Table 3. Task 1: F1 scores by language (Test set)

Team	F1-en	F1-ru	F1-de	F1-fr
ACSS-PSL	0.7208	0.5379	0.6878	0.7374
Mean	0.5915	0.4721	0.5567	0.5876
Median	0.6924	0.5366	0.6686	0.6865

Table 4. Task 1: Macro-averaged overall results (Test set)

Team	F1-score	Precision	Recall
ACSS-PSL	0.6333	0.6222	0.6449
Mean	0.5394	0.5440	0.5661
Median	0.6268	0.6172	0.6312

Task 6: Overall F1-Score of 0.951

Our system achieved a final F1-score of 0.951, slightly surpassing the official baseline. The ensembling strategy provided a clear benefit over a single model (0.951 vs 0.942).

Table 5. Task 6: Final results (Test set)

Team	F1-score	Precision	Recall
ACSS-PSL	0.951	0.940	0.962
Mean	0.938	0.916	0.961
Median	0.944	0.922	0.972

Acknowledgements

We thank the SMM4H 2025 organizers for providing the datasets and evaluation framework.